Computer Science Department
Template for Updated Requirements and Project Plan – Spring 2026

Title of the project:	"Smart Campus Navigation with Real-Time Positioning and Adaptive Accessibility Routing"
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Requirements (50%)

1. Stakeholders

The Smart Campus Navigation app involves multiple stakeholders who either use it directly or are affected by its working:

- **Students (Primary Stakeholders):** Undergraduate and graduate students use the system for its navigation purposes, accessibility-aware routing, planning, event tracking, discounts provided by MEA, and personalized recommendations of study rooms.
- **New Students and Campus Visitors:** Freshmen, exchange students, parents of any student, and visitors who will use the app for navigation and orientation on campus.
- **Students with Disabilities:** Socially vulnerable students, users who are dependent on accessibility features of the system (elevators, ramps, tactile paths, barrier-free navigation)
- **University Administration:** The departments of University Administration that provide student services enhance digital infrastructure, which will develop campus accessibility, student engagement, and information delivery.
- **IT and Technical Support Staff:** The university's IT Department is engaged in system integration, service provision, data security and privacy, and sustainable maintenance.
- **Campus Service Providers:** Service providers, such as cafes, libraries, study rooms, and other facilities, whose information, reviews, and other data will be displayed to the users.
- **External Vendors and Partners:** External entities providing special discounts for students that integrate with the platform through APIs or other features.
- **Project Advisors and Faculty:** Academic supervisors, professors, or teachers' assistants who evaluate the system's design, its feasibility, and compliance with project requirements.

2. Functional System Requirements

1. **Positioning and Localization**
    - 1.1 The system shall provide outdoor GPS positioning with an accuracy of approximately ±3 meters.
    - 1.2 The system shall provide indoor positioning using Wi-Fi triangulation and Bluetooth beacons with an accuracy of 1-2 meters.
    - 1.3 The system shall automatically and seamlessly transition between indoor and outdoor positioning modes without requiring user intervention.
    - 1.4 The system shall provide a fallback positioning mechanism using Wi-Fi fingerprinting and device sensors (accelerometers, gyroscopes) with ≤5m accuracy if beacons are unavailable.
2. **Adaptive Routing**

- 2.1 The system shall support multi-criteria routing, allowing users to select the shortest, most accessible, or least crowded paths.
- 2.2 The system shall identify and integrate accessibility metadata, including elevators, ramps, tactile paths, and steep inclines into route calculation.
- 2.3 The system shall perform real-time rerouting based on dynamic campus conditions such as construction, event occurrences, and crowd density.
- 2.4 The system shall provide offline-first navigation using cached maps and pre-computed routes for use during network instability or emergencies.

3. **Augmented Reality (AR) Navigation**
    - 3.1 The system shall provide directional navigation using AR-based arrows, markers, and landmark recognition at a stable 60 FPS.
    - 3.2 The system shall automatically switch to a 2D map interface for devices that do not support ARCore or ARKit.
4. **Campus Services and Social Features**
    - 4.1 The system shall include an academic planner and GPA calculator synchronized with university registrar data.
    - 4.2 The system shall provide an AI-powered review system that uses Natural Language Processing (NLP) to perform sentiment analysis on user feedback for campus facilities.
    - 4.3 The system shall offer a student discount and rewards platform, verifying eligibility and awarding loyalty points for app engagement.
    - 4.4 The system shall provide personalized recommendations for study spots and dining based on user history and real-time noise or crowd levels.
5. **Security and User Management**
    - 5.1 The system shall support secure user authentication via OAuth 2.0 and JWT token-based sessions.
    - 5.2 The system shall provide an "Anonymous Mode" to allow basic navigation without tracking personal identity.

3. Non-functional Requirements

- **Performance:** Route calculations and review submissions must be completed in less than 2 seconds.
- **Scalability:** The system must support at least 1,000 concurrent users during peak hours without performance degradation.
- **Availability:** The platform aims for a 99.5% uptime with automated failover mechanisms in place.
- **Usability:** The interface must comply with WCAG 2.1 Level AA accessibility standards, including support for high-contrast modes, scalable text, and audio guidance.
- **Security & Privacy:** The system must ensure end-to-end encryption for all sensitive data and comply with GDPR and FERPA regulations.
- **Reliability:** The sentiment analysis model for user reviews must maintain a precision and recall rate of over 85%.
- **Compatibility:** The mobile application must support iOS 13+ and Android 8.0+ through a cross-platform framework (Flutter or React Native).

4. Domain Requirements.

These requirements have been developed based on the university regulations, characteristics of the university infrastructure, and educational data governance.

- **Campus Mapping and Infrastructure Constraints:**

  It also needs to ensure maximum operation using approved campus maps issued by the university administration. All buildings, pathways, entrances, elevators, ramps, and restricted areas must depict the actual physical campus layout. The navigation routes must avoid staff-in-access zones, emergency exits, and construction zones unless allowed to access them.

- **Accessibility and Inclusion Standards:**

  The system also has to adhere to the accessibility policies applicable to higher education institutions. The routes marked “accessible” must include only routes known to be accessible; i.e., elevator routes must be working, ramps must be within acceptable inclines, and the tactile route must be available. The data must be updated to include outages, e.g., elevator outages.

- **Academic Data Integrity:**

  Any feature that is academic-related (such as a GPA calculator, academic planner, event schedule, etc.) must be able to use data from official university sources/APIs. The system shall not allow manual editing of official academic records. The system shall run as read-only against registrar data.

- **Privacy, Consent, and Educational Regulations:**

  The system must adhere to the GDPR and FERPA guidelines on student data. Location tracking, personalisation, and analysis can be done based on user consent. An anonymous mode must be implemented such that identity-related data storage is disabled, while basic navigation functionality is retained.

- **Operational Reliability in Campus Conditions:**

  The system will have to cope with the particular operational conditions which are typical of a campus, including non-constant wireless availability, heavy user density during transitions between classes, and environmental changes that are temporary. Offline routing will have to support basic routing between significant campus points.

- **Integration with Campus Services:**

  The system must accommodate a controlled integration with campus service providers such as libraries, study rooms, cafeterias, etc., along with external entities using standardised APIs. The integration must conform to university's information technology security policies regarding third-party integrations.

5. Assumptions and Justifications

The development of the Smart Campus Navigation system is based on the following assumptions, which, in turn, affect design decisions and the system’s scope:

- **Availability of Campus Infrastructure:**
  The main assumption is that the campus provides stable GPS coverage outdoors and sufficient Wi-Fi coverage indoors.
- **User Device Capabilities:**
  One of the other assumptions is that users own modern smartphones that can download the app

and support GPS Wi-fi. However, not all devices support ARCore or ARKit, which justifies the inclusion of a 2D map fallback.

- **Access to University Data Sources:**
  Our system assumes that there is controlled access to university services, such as authentication during login, academic schedules, and event data that are synchronized through APIs.
- **User Consent and Privacy Awareness:**
  Users will explicitly grant permission for their location and other personalized features. However, to avoid privacy concerns, an anonymous mode may be used for basic navigation without identity tracking.
- **Limited Real-World Training Data:**
  Machine Learning components, such as sentiment analysis and crowd prediction, will be trained on limited or synthetic datasets. Thus, the accuracy will increase over time since the data will be more available.
- **Network Connectivity Is Not Guaranteed:**
  There might be cases when users experience unstable network conditions; thus, the system assumes intermittent connectivity and switches to an offline-first design with already saved maps and pre-calculated routes.
- **Scalability Needs Will Grow Gradually:**
  Initial deployment targets a controlled user base (pilot scale), with the assumption that cloud-based infrastructure can be scaled as adoption increases.

The assumptions above are written based on technical limitations, privacy and security regulations, and the phased nature of implementation, making sure that the system will be usable, secure, and extended within the academic environment.

Project Management and Tasking (50%)

1. High-Level Tasking Plan for Spring Semester

T*he Spring semester will focus on full system implementation, integration, testing, and deployment readiness.*

- *System Architecture Finalization*
  - *Finalize backend architecture (microservices, routing engine, authentication flow)*
  - *Confirm data models for routing, accessibility metadata, reviews, and events*
  - *Define API contracts between backend and mobile application*
- *Backend Development*
  - *Implement routing engine (multi-criteria routing: distance, accessibility, crowd density)*
  - *Integrate hybrid positioning logic (GPS + simulated indoor positioning)*
  - *Implement ML services (sentiment analysis for reviews, basic crowd prediction)*
  - *Optimize database queries and indexing*
  - *Integrate university services (academic schedule, events, authentication)*
- *Mobile Application Development*
  - *Complete navigation module (2D + AR fallback)*
  - *Implement accessibility-aware routing and UI options*
  - *Finalize academic tools (GPA calculator, planner)*
  - *Complete campus services (events, discounts, reviews)*
  - *Cross-platform testing (iOS & Android)*
- *Integration & System Testing*
  - *Backend ↔ mobile API integration*
  - *End-to-end functional testing*
  - *Accessibility testing (WCAG-oriented checks)*

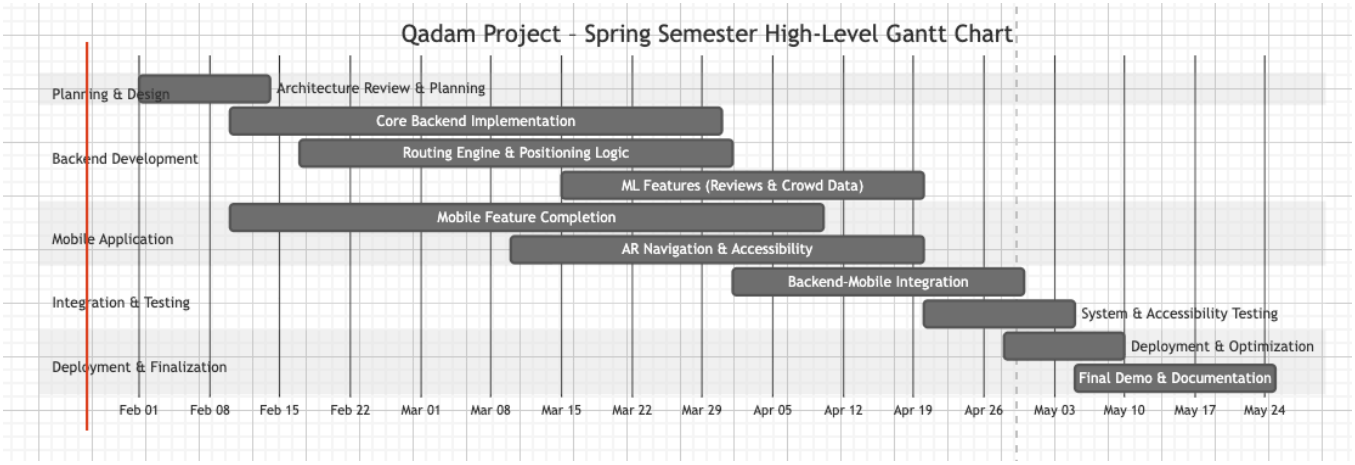
    - *Performance and load testing*
- *Deployment & Evaluation*
    - *Deploy backend services to cloud environment*
    - *Internal pilot testing with sample users*
    - *Bug fixing and performance tuning*
    - *Prepare final demo and documentation*

1.1 Project Management Tools Used

The team continues using Agile methodology with weekly sprints. Tools:

- *Notion for sprint planning, task tracking (Kanban boards)*
- *GitHub for version control and collaboration*
- *Figma for UI/UX design updates*
- *Docker for backend service containerization*
- *Gantt.com for high-level project timeline visualization*

2. Gantt Chart



3. Team responsibilities and task allocation

To ensure efficient development and clear ownership, responsibilities for the Spring semester are distributed among team members based on technical focus and prior contributions.

- **Aiym Abay** is responsible for backend architecture and core system logic. This includes finalizing the microservices structure, implementing the multi-criteria routing engine, integrating positioning logic, and optimizing database performance. Aiym will also support backend–mobile API integration and assist with deployment preparation.
- **Zhansaya Zhassulanova** focuses on mobile application development and UI/UX implementation. Her tasks include completing the navigation interface (2D map and AR fallback), implementing accessibility-aware UI features such as high-contrast mode and scalable text, and integrating backend APIs into the mobile application using Flutter.
- **Aruzhan Ryssayeva** is responsible for augmented reality features and accessibility routing logic. This includes AR-based navigation implementation, landmark guidance, testing AR performance, and ensuring that accessible routes (elevators, ramps, barrier-free paths) are correctly visualized and functional.
- **Aiymzhan Ashirbek** is responsible for academic tools and system integration. Tasks include implementing the GPA calculator, academic planner, synchronization with calendar and event data, and ensuring read-only integration with university academic information sources.

Aiymzhan will also assist with system testing and documentation.

- **Madina Dairova** leads machine learning features and social functionality. Her responsibilities include implementing sentiment analysis for user reviews, basic crowd prediction logic, review moderation mechanisms, and personalized recommendation features. She will also assist with performance testing and evaluation of ML components.

All team members will jointly participate in system testing, bug fixing, documentation preparation, and final demo presentation.

4. Expected outcomes by the end of the Spring Semester

By the end of the Spring semester, the project will deliver a *fully integrated and tested functional* prototype of the Smart Campus Navigation application.

The backend system will include a working routing engine with multi-criteria optimization (distance, accessibility, and crowd awareness), secure authentication, academic data integration, and ML-based review analysis. Core APIs will be deployed in a cloud environment and connected to the mobile application.

The mobile application will support both iOS and Android, providing indoor and outdoor navigation with accessibility-aware routing, 2D maps with AR navigation fallback, academic tools such as a GPA calculator and planner, campus event listings, discount services, and a review system.

Accessibility features will be fully implemented, including high-contrast mode, scalable text, accessible route filtering, and offline emergency routing. The system will comply with WCAG 2.1 AA guidelines and privacy-by-design principles, including an anonymous navigation mode.

Comprehensive testing will be completed, including functional testing, accessibility checks, performance evaluation, and end-to-end integration testing. The final deliverables will include a live demo, system documentation, updated requirements, and a deployment-ready prototype suitable for pilot use on campus.